Entrance Examination for the Department of Electrical and Electronic Engineering, Graduate School of Engineering, University of Tokyo

Physics

12:30 ~ 15:30, Tuesday, August 28, 2007

Instructions

1. Do not open this booklet before permission is given.

2. This booklet contains eight problems. If you find missing or badly printed pages, ask the proctor for exchange.

3. Fill in the top parts of all your four answer sheets and scratch paper. Before submitting your answer sheets, make sure that the top parts are correctly filled in.

4. Answer four problems out of the eight. Your answer to each problem should be written on a separate sheet. You may use the reverse side of the sheet if necessary.

5. The four answer sheets must be submitted at the end of the examination, even if they are blank ones. (The top parts of all the four sheets must be filled in.)

6. You must answer in Japanese or English.

7. This booklet and the scratch paper must be returned at the end of the examination.

8. Problems in Japanese are regarded as official ones. This English translation is provided for the convenience of applicants.

Applicant Number	

Write your applicant number in the above box.

Problem 1

A coaxial capacitor of length L is formed by two conducting cylinders A and B of radii a and b $(a < b)$, respectively, as shown in Fig. 1. The region between the radii a and d $(a < d < b)$ is filled with a dielectric of permittivity ε_1, and the region between the radii d and b is filled with a dielectric of permittivity $\varepsilon_2\ (< \varepsilon_1)$. Neglecting edge effects, answer the following questions.

(1) Switch SW is closed and the capacitor is charged to potential V_0. Find the electric field strength at distance r from the cylindrical axis and determine the capacitance C of the capacitor.

(2) Find the radius a such that the electric field strength on the outer surface of cylinder A is a minimum for fixed radii b and d.

(3) Switch SW is opened after the capacitor is charged to potential V_0. Find the magnitude and direction of electrostatic force per unit area on cylinder B using the principle of virtual work.

(4) In (3), find the magnitude and direction of resultant force on a part of cylinder B subtending a central angle $\theta\ (< \pi)$ (shaded area shown in Fig. 2).

(5) Switch SW is closed again. Find the magnitude and direction of electrostatic force per unit area on cylinder B using the principle of virtual work.

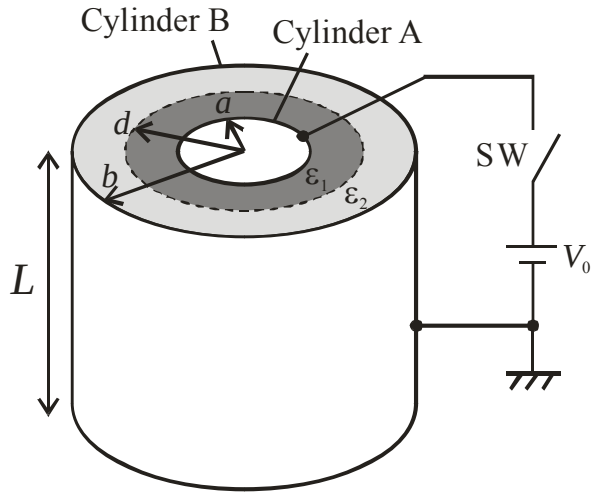


Fig. 1

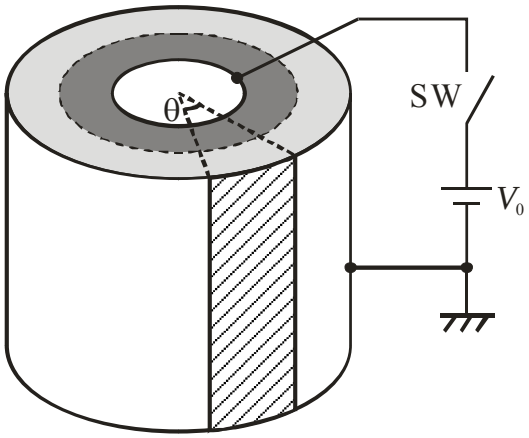


Fig. 2

Problem 2

Answer the following questions by considering an ac electric field at a frequency f in the material (electric conductivity: σ, permittivity: ε, permeability: μ). In the following questions, $\boldsymbol{J}_c$ and $\boldsymbol{J}_d$ represent the conduction current density and the displacement current density, respectively.

(1) Find the condition of frequency f where $|\boldsymbol{J}_d| << |\boldsymbol{J}_c|$.

(2) Compare $\boldsymbol{J}_d$ and $\boldsymbol{J}_c$ at f = 1GHz in the materials listed in the following table. Here, free space permittivity ε_0 is 9×10^{-12} F/m.

	Alumina	Seawater	Copper
Electric conductivity[S/m]	1×10^{-10}	2	5×10^{7}
Relative permittivity	8	80	1

(3) Answer the following questions under the condition $|\boldsymbol{J}_d| << |\boldsymbol{J}_c|$. If necessary, you may use following equations ($\boldsymbol{X}$: vector).

$$\mathrm{div}\cdot\mathrm{rot}\,\boldsymbol{X} = 0$$

$$\mathrm{rot}\cdot\mathrm{rot}\,\boldsymbol{X} = \mathrm{grad}\cdot\mathrm{div}\,\boldsymbol{X} - \nabla^2\boldsymbol{X}$$

$$\nabla^2\boldsymbol{X} = \nabla^2\left(X_x,\ X_y,\ X_z\right) = \left(\nabla^2 X_x,\ \nabla^2 X_y,\ \nabla^2 X_z\right)$$

(3-1) Show $\mathrm{div}\,\boldsymbol{D} \cong 0$ ($\boldsymbol{D}$: electric flux density).

(3-2) Show $\nabla^2\boldsymbol{J}_c \cong \sigma\mu\dfrac{\partial \boldsymbol{J}_c}{\partial t}$.

(3-3) Suppose that an ac current at an angular frequency ω flows in the cylindrical conductor made by the uniform material of radius a, and that this current is symmetrical in parallel with the axis of the conductor. In cylindrical coordinates (r, θ, z) along the conductor, the conduction current density $\boldsymbol{J}(r,t) = \boldsymbol{R}(r)e^{j\omega t}$ meets the equation (*).

$$\nabla^2\boldsymbol{J}(r,t) = \frac{\partial^2\boldsymbol{J}(r,t)}{\partial r^2} + \frac{1}{r}\frac{\partial\boldsymbol{J}(r,t)}{\partial r} \qquad (*)$$

When the second term in the right-hand side of eq. (*) is negligibly small, find the amplitude $|R_z(r)|$ $(0 \le r \le a)$ under the boundary condition $|R_z(a)| = R_{z0}$ and $|R_z(0)| < R_{z0}$.

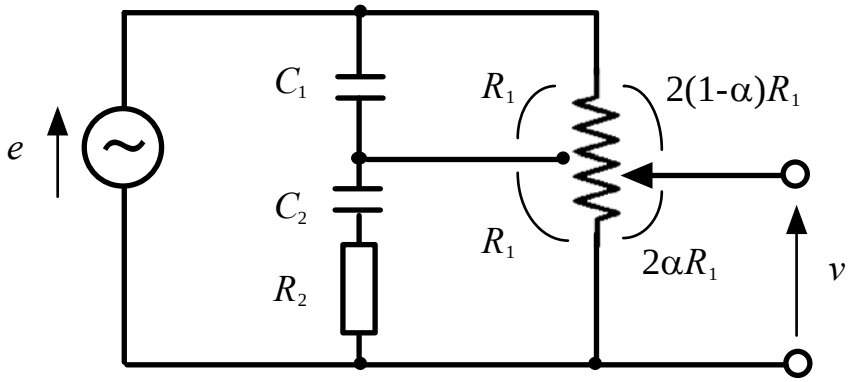
Problem 3

The figure shows the loudness circuit which emphasizes lower and higher frequencies, when the volume of an audio amplifier is squeezed. The input signal e is output as v. α ($0 \leq \alpha \leq 1$) shows the position of the output terminal of the variable resistor with a center tap. The impedance of the input signal source is small enough.

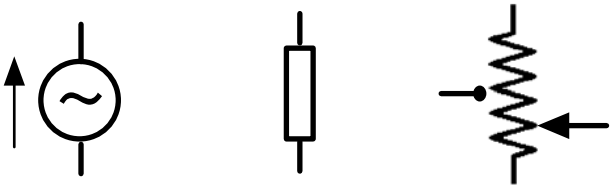
(1) When $\alpha = 0.5$, derive the transfer function from e to v.

(2) With suitable approximation, give the factorizations of the denominator and nominator of the transfer function derived in (1) , and illustrate the rough shape of its frequency characteristics of the amplitude by using polygonal line approximation. When making approximation, use $R_1 = 50\ \mathrm{k\Omega}$, $R_2 = 1\ \mathrm{k\Omega}$, $C_1 = 1000\ \mathrm{pF}$, $C_2 = 0.1\ \mu\mathrm{F}$ as an example, and utilize the relation that C_2R_1 is much larger than C_2R_2, C_1R_1 and C_1R_2.

(3) When α varies from 0 to 1, explain the change of the frequency characteristics mentioned above by drawing figures.



note:



voltage source | resistor | variable resistor with center-tap

Problem 4

Figure 1 shows a field effect transistor (FET) that has three electrodes of G, D and S. Figure 2 shows a small signal (ac) equivalent circuit of the FET. When a small ac voltage v_{gs} is applied between G and S, a current $g_m v_{gs}$ flows from D to S. A resistance between D and S is r_d.

(1) Convert the small signal equivalent circuit in Fig. 2 to a small signal equivalent circuit using a voltage source.

(2) Figure 3 shows a circuit with two FETs and resistances that have two inputs and two outputs. FETs T_1 and T_2 have the same characteristics. The small ac voltages of two inputs are v_1 and v_2. The dc voltages V_1 and V_2 of two inputs are the same. The small ac voltages of two outputs are v_3 and v_4. V_{DD} is the supply voltage. Show the small signal equivalent circuit of this circuit in Fig. 3 using the small signal equivalent circuit obtained in (1).

(3) Assume that the difference of the two inputs ($v_1 - v_2$) is v_{in} and that the difference of the two outputs ($v_3 - v_4$) is v_{out}.

(3-1) What is the name of the circuit operating at this condition?

(3-2) Express v_{out} using v_{in} and find the voltage gain A ($=v_{out}/v_{in}$).

(3-3) Show schematically a relation between v_{out} and v_{in} in a graph with the vertical axis of v_{out} and horizontal axis of v_{in}.

(4) Assume $v_{in} = v_1 + v_2$ and $v_{out} = v_3 + v_4$. Find the voltage gain A' ($= v_{out}/v_{in}$).

(5) Compare A and A'.

(6) Considering the results of (3)–(5), what applications is this circuit suitable for? Show an example and describe the reason.

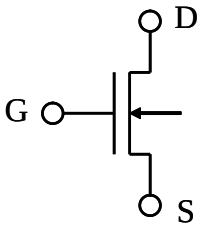


Fig. 1

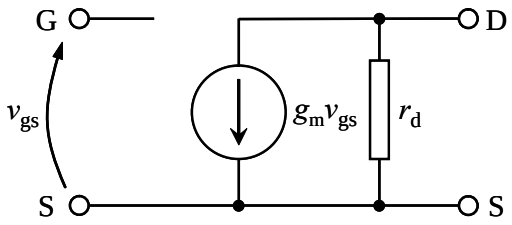


Fig. 2

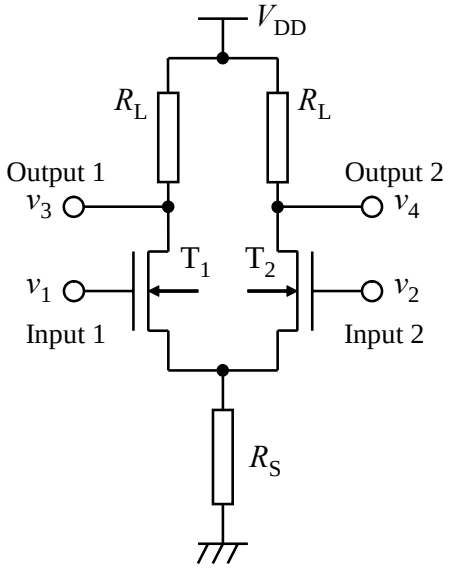


Fig. 3

Problem 5

(1) Let us consider a homogeneous rigid circular plate shown in Fig.1 whose radius is *a*, mass is *M*, and thickness is negligibly small. Define the $X_bY_bZ_b$ Cartesian coordinate system whose origin is the center of the plate, Z_b-axis is perpendicular to the plate, X_b-axis and Y_b-axis are on the plate. Calculate the moment of inertia around Z_b-axis.

(2) Prove $I_Z = I_X + I_Y$ where I_X, I_Y and I_Z are the moments of inertia around X_b-, Y_b- and Z_b-axis in Fig.1, respectively.

(3) Calculate the moment of inertia of the circular plate around the Z-axis which is parallel to Z_b-axis at a distance of *h* as shown in Fig. 2.

(4) Let us consider that the plate in Fig.1 is connected with a negligibly-small-mass spring to the origin of XYZ coordinates, as shown in Fig.3. At first, the spring and the plate were rotating together around Z-axis with the angular velocity of ω_0, and the spring was restricted to its natural length L_0. After the restriction had released, the spring started elastic oscillation. Finally, due to kinetic energy loss of elastic motion of the spring, the length of the spring became L_1 and the rotational speed became ω_1. Calculate the ratio of ω_1 to ω_0, when $L_0=2a$ and $L_1=3a$. Assume that one end of the spring is connected to the origin so as to rotate around Z-axis without friction, and another end of the spring is connected to the circumference of the plate.

(5) In (4), calculate the kinetic energy loss caused by the elastic motion of the spring, using ω_1, *M* and *a*.

(6) Let us consider that the spring is cut when the plate was rotating with the angular velocity of ω_1 in (4). Express the translational and rotational motions of the plate, using $\omega_1$and *a*.

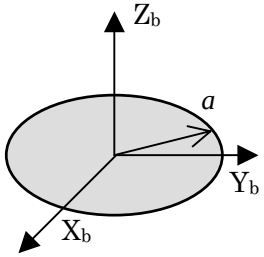


Fig.1

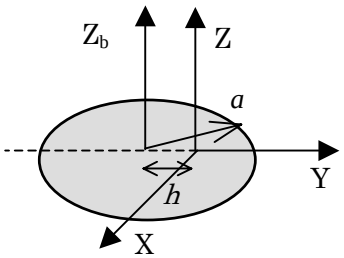


Fig.2

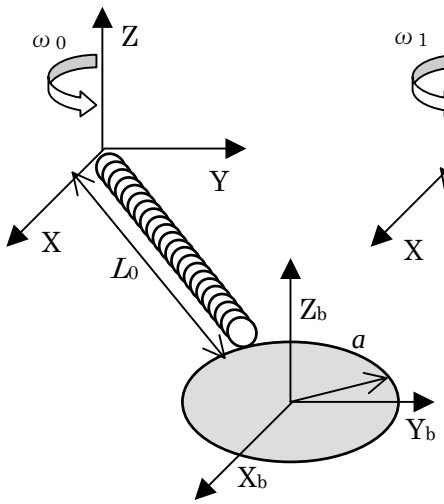


Fig.3

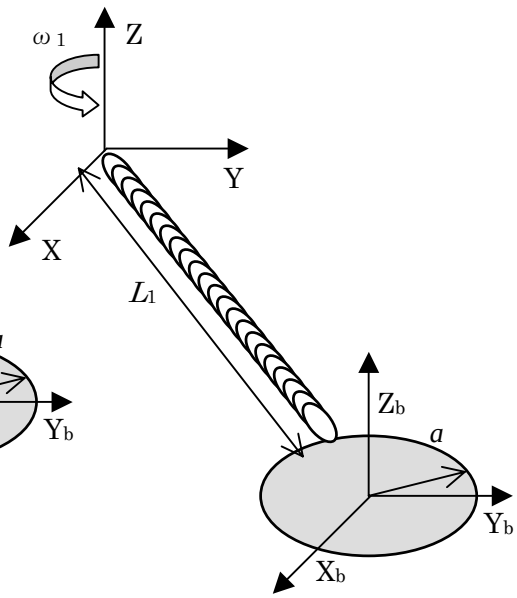


Fig.4

Problem 6

Consider gas molecules with the number of N filled in a vessel with a size of L cube in a condition of the normal state. Each gas molecule with a weight of m is moving with a velocity of v. Their velocity distribution is given by $f(v)$. An impulse generated by a collision of gas molecule to a wall causes a pressure P. If a molecule impacts a wall in an x-axis direction with a velocity of v_x under the perfectly elastic collision condition, the impulse of one molecule is given by

$\Delta(mv_x) =$ [A] .

A frequency of the collision λ to the same wall is

$\lambda =$ [B] .

The impulse given by a molecule to a wall during a period of Δt is given by

[C] .

In an actual condition, each velocity of molecule changes at every collision, but the overall velocity distribution $f(v)$ doesn't change. The total impulse of all molecules is given as a product of an average velocity and the number of molecules N. This is an observed macroscopic pressure P, which is given by

$P =$ [D] ,

where $\overline{v_x^2}$ is an average value of v_x^2.

(1) Fill the blanks of A～D.

(2) Express a root mean square velocity of gas molecules whose total mass is M.

(3) Assuming N_2 molecule as a gas with a concentration of 1 mole, calculate its root mean square velocity. Here, 1 atm is 1.013×10^5 N/m^2, and a mass of 1-mole N_2 gas is 0.028kg.

(4) Actual diffusion velocity of a gas molecule is less than 1m/s in general. This value is quite smaller than that given in (3) for the N_2 gas molecule which is assumed to be a sphere with a diameter of 3nm. Explain its reason numerically.

Problem 7

Consider a dielectric waveguide with a core layer (refractive index: n_1, thickness: a) and clad layers (refractive index: n_2) as shown in Fig. 1, where $n_2 < n_1$. There is no variation of the refractive index in the *y* direction. The *y*-polarized electromagnetic wave propagating to the *z* direction, $E(x,z) = E_y(x)\exp j(\omega t - \beta z)$, satisfies the following equation:

$$\frac{d^2E_y}{dx^2} + \left[k_0{}^2 n^2(x) - \beta^2\right]E_y = 0 .$$

$k_0 (= 2\pi/\lambda_0 = \omega/c)$ is a wavenumber in vacuum, $n(x)$ is a refractive index, β is a propagation constant, λ_0 is a wavelength in vacuum, ω is an angular frequency, c is light velocity in vacuum, and *j* is an imaginary unit.
Answer the following questions.

(1) When the electrical fields $E_y(x)$ in the core and clad are given by

$$E_y = \exp jk_1 x \text{, in the core, and}$$
$$E_y = \exp jk_2 x \text{, in the clad,}$$

express k_1 and k_2 using β.

(2) For the guide modes, k_1 and k_2 should be a real number and an imaginary number, respectively. Derive the following condition necessary for the guide modes,

$$n_2 < n_{eff} < n_1$$

where $n_{eff} (\equiv \beta/k_0)$ is an effective refractive index.

(3) Assume that the electromagnetic wave is well confined in the core region, i.e. $E_y(x) = 0$ in the clad region. Derive n_{eff} of the guide modes, and draw $E_y(x)$ along the *x*-axis for the three guide modes in descending order of n_{eff}.

(4) The electromagnetic wave is hardly guided when n_{eff} derived in (3) is smaller than n_2. Calculate the thickness condition of *a* for a single mode waveguide (i.e. there is only one guide mode) by using $n_1 = 3.0$, $n_2 = 1.0$, and $\lambda_0 = 1.55$ μm.

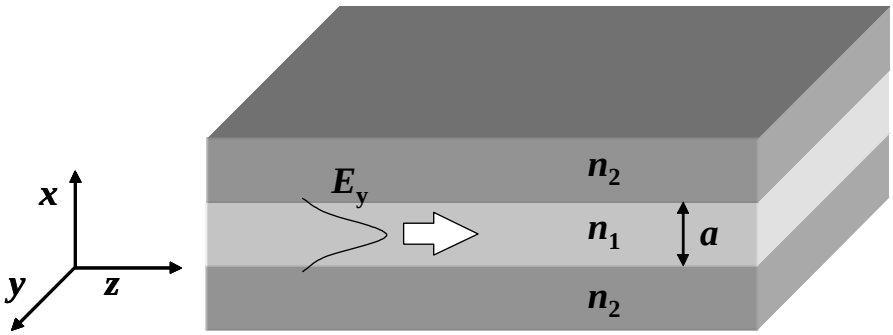


Fig. 1

Problem 8

We consider a one-dimensional harmonic oscillator, which is oscillating at a frequency of f. When this harmonic oscillator is in the n-th quantum mechanical energy level, the energy eigenvalue E_n is expressed as $\boxed{E_n = \text{(A)}}$ by using n, f, and the Planck's constant h. Note that the ground state is specified by $n = 0$.

Assume that this harmonic oscillator is in thermal equilibrium at absolute temperature of T. In such a case, the harmonic oscillator occupies the n-th energy level with probability of P_n. Similarly, the harmonic oscillator occupies the m-th energy level with probability of P_m. The ratio of these probabilities P_n / P_m is expressed as $\boxed{P_n / P_m = \text{(B)}}$ by using E_n, E_m, T, and the Boltzmann constant k.

(1) Show expressions for the terms (A) and (B).

(2) Calculate the probability P_n by using $\sum_{n=0}^{\infty} P_n = 1$.

(3) Calculate the averaged energy $\bar{E}$ of the harmonic oscillator by using the derived expression for P_n in (2). You may use the following formula if necessary.

$$\sum_{n=1}^{\infty} n x^{n-1} = \frac{1}{(1-x)^2} \quad (|x| < 1)$$

(4) From the result of (3), obtain the averaged energy $\bar{E}$ when $hf / kT << 1$.

(5) Show that the result of (4) can be obtained by using the classical statistical mechanics for the harmonic oscillator.